THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS CHAPTER TEST SET B

Chapter 3: **A Story of Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The Roman numeral XCIV stands for: 1

(a) 95 **(b)** 94 **(c)** 104 **(d)** 84

2. The Roman numeral MMXXIV stands for: 1

(a) 2014 **(b)** 2024 **(c)** 2034 **(d)** 2025

3. 400 in Roman numerals is: 1

(a) CDI **(b)** CDX **(c)** CD **(d)** CCCXCIX

4. The decimal number 17 written in base 3 is: 1

(a) 122 **(b)** 201 **(c)** 200 **(d)** 121

5. The decimal number 100 written in base 7 is: 1

(a) 202 **(b)** 203 **(c)** 204 **(d)** 201

6. The value of 1000 in base 4, in the decimal system, is: 1

(a) 64 **(b)** 66 **(c)** 63 **(d)** 65

7. The first known written use of the digit zero as a dot is in the: 1

(a) Lilavati **(b)** Arthashastra

(c) Rigveda **(d)** Bakhshali manuscript

8. The Mesopotamian number system was based on: 1

(a) 10 **(b)** 60 **(c)** 20 **(d)** 5

9. The idea of mapping each object to exactly one stick or pebble is called: 1

(a) base conversion **(b)** one-to-one mapping

(c) place value **(d)** rounding

10. The Roman system does not have a symbol for: 1

(a) five **(b)** zero **(c)** fifty **(d)** one thousand

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Convert 1011_2 to decimal. 2

12. Convert 19 to base 2. 2

13. Why is the Hindu number system considered a great invention? 2

14. Name two ancient civilisations that had their own number systems. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Convert 45 to base 2 showing the steps. 3

16. Convert 60 to base 3 showing the steps. 3

17. Convert 88 to base 5 showing the steps. 3

OR

Convert 123 to base 6 showing the steps. 3

18. Convert 200 to base 8 showing the steps. 3

OR

Convert 77 to base 4 showing the steps. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Describe the contribution of Aryabhata and Al-Khwarizmi to numbers. 5
(ii) Why do we say 'Hindu–Arabic numerals'?

OR

(i) Write the expanded form of 2304 in base 10. 5
(ii) Write 2304_5 in decimal.

20. **Case study:** Computers store data in binary. A byte has 8 binary digits. 5
(i) How many different values can a byte store?
(ii) Write the largest value in decimal.
(iii) Convert 200 to binary.

OR

(i) Subtract XLVII from C in Roman numerals. 5
(ii) Write the answer in Roman numerals.

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND